ANURAG MISHRA

Full-Stack Engineer · AI/ML Researcher · Edge Computing & Vision Systems

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PROFESSIONAL SUMMARY

B.Tech Information Technology student and full-stack software engineer with hands-on experience in JavaScript, TypeScript, Python, React.js, Node.js, and MongoDB. Currently building real-time computer vision pipelines on NVIDIA Jetson AGX using DeepStream and PyQt5 at C-DAC Thiruvananthapuram. Co-authored and presented an IEEE-format research paper on AI-based livestock monitoring at an international conference. Proven track record in production deployments, distributed systems, and cross-functional team leadership.

EDUCATION

B.Tech, Information Technology

Jul 2023 – Jul 2027 (Expected)

Dr. Ram Manohar Lohia Avadh University · Ayodhya, India

CGPA: 7.22 (Yr 1) · 7.05 (Yr 2) · 7.12 (Yr 3)

Relevant Coursework: Data Structures & Algorithms, Machine Learning / AI, Linear Algebra, Probability & Statistics, DBMS, Operating Systems, Object-Oriented Programming, Software Engineering, System Design

RESEARCH & PROFESSIONAL EXPERIENCE

Software Trainee – Work-Based Learning (WBL)

Aug 2026 – Feb 2027

C-DAC Thiruvananthapuram · Thiruvananthapuram, India

- **Engineering real-time computer vision pipelines** on **NVIDIA Jetson AGX** using **NVIDIA DeepStream SDK** and **GStreamer**, ingesting live camera RTSP streams for mission-critical surveillance.
- **Built a PyQt5-based GUI streamer** to display and control multi-channel video feeds from DeepStream pipelines, enabling operators to monitor and interact with live inference output.
- **Building and maintaining real-time distributed monitoring dashboards** (Grafana, Prometheus) tracking system health and application-level metrics on live production infrastructure.
- Gained deep hands-on exposure to **fault-tolerant system design, CUDA-accelerated inference, edge AI deployment, and observability pipelines** in a zero-downtime production environment.

AI Research Intern – Full-Stack & Data Engineering

May 2026 – Jul 2026

IIIT Naya Raipur · Naya Raipur, India

- Conducted structured critical analysis of **CowNet-AI** (published IEEE Access multi-agent system), identifying **10 specific deployment & architectural gaps** against a benchmark dataset.
- **Co-authored IEEE-format research paper** proposing edge-cloud hybrid architecture with temporal graph transformers, multimodal sensor fusion, SHAP/LIME explainability, and federated learning.
- **Presented at ICNACT 2026** (3rd Intl Conference on Nonlinear Analysis & Computational Techniques), VIT Bhopal.
- Designed & shipped a **multi-tiered full-stack platform** (Next.js, Node.js, MongoDB Atlas) end-to-end, integrating third-party APIs with graceful fallback under tight deadlines.

Team Manager – CDO Intern (Full-Stack)

Apr 2025 – Jun 2025

Chief Development Office, Ayodhya · Govt. of UP · Ayodhya, India

- **Led a cross-functional team** to design, build, and deploy a full-stack **SHG Inventory Management System** end-to-end: live inventory tracking, sales/profit monitoring, and QR-code data ingestion.
- Implemented **role-based access control**, live stock updates, and QR-code stock verification, cutting manual entry errors for end users across multiple Self Help Groups.

PROJECTS

HerdIntel – Precision Dairy Intelligence Platform | Live Demo

React.js · Node.js · Express.js · MongoDB · Chart.js

- Full-stack MERN platform for real-time dairy cattle welfare monitoring — **deployed live**; dashboards visualise isolation risk scores, real-time alerts, and social-network graphs for non-technical, rural end users.
- Backed by **co-authored IEEE research paper** identifying 10 deployment gaps in an existing multi-agent system and proposing temporal graph transformer + SHAP-explainable + federated-learning architecture; **presented at ICNACT 2026**.

Ayodhya Blessings – The Divine of Ayodhya | Live Demo

React.js (TypeScript) · Node.js · Express.js · Vite

- Built and deployed a full-stack TypeScript web app with **decoupled React (Vite) frontend** and Node.js/Express REST API backend; independently developed, version-controlled, and shipped to production.

React E-Commerce MERN Application

React (TypeScript) · Node.js · Express.js · MongoDB · Redux

- **JWT authentication**, password hashing, shopping cart, order management, and payment gateway integration; deployed via CI/CD with Redux-managed global state.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, SQL, HTML/CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Flask, Redux, Context API, Tailwind CSS, PyQt5

ML / AI: scikit-learn, NumPy, pandas, Matplotlib; PyTorch & TensorFlow (exposure); LLM/Gemini API; Federated Learning

Edge AI & Vision: NVIDIA DeepStream SDK, GStreamer (pipeline design), CUDA (GPU inference), NVIDIA Jetson AGX, RTSP stream processing, Real-time computer vision pipelines

Databases: MongoDB (NoSQL), MySQL, SQLite, SQL query optimisation

Cloud & DevOps: Git/GitHub, Vercel, CI/CD, Docker, Linux, Grafana, Prometheus, AWS (Forage simulation)

Core CS: DSA (Arrays, Trees, Graphs, DP, BFS/DFS), REST API design, System Design, Distributed Systems, Fault-tolerant design, OOP, Agile/Scrum

CERTIFICATIONS & ACHIEVEMENTS

- **Conference Presentation** – ICNACT 2026, VIT Bhopal University · Paper: "HerdIntel: A Web-Based Platform for Real-Time Dairy Cow Welfare Monitoring"
- **Research Internship Certificate** – IIIT Naya Raipur Outreach Program
- **AWS Solutions Architecture** – Forage Job Simulation
- **Introduction to Strategy Consulting** – BCG, Forage | **Operations Industrial Engineer** – Siemens, Forage
- **CDO Intern Participation Certificate** – Chief Development Office, Ayodhya
- CodeAlpha Participation Certificate | Olympiad Participation Certificate